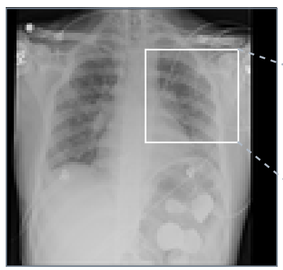
(a) Reading a concept is not the same as controlling it



Chest radiograph

Split into patches

Vision tower

Image → visual tokens

Hatched: excluded

READ · Linear probe → Concept score

Lift $\beta_c \rightarrow \hat{w}_c$

Mean pool → project → standardize

Consumed visual tokens h_{it}

Written tokens h'_{it}

Same token positions

WRITE · Add a direction

$$h'_{it} = h_{it} + \alpha \, \|h_{it}\|_2 \, \hat{w}_c$$

$\alpha = +0.25$

Is there a pleural effusion in this chest radiograph? Answer yes or no.

Connector

Maps visual tokens to language model

Language model

Answers the question using written tokens

$P(\text{yes} \mid q)$

Writing changes feature activations, not the radiograph.

(b) Ownership: does the concept's own direction win?

Same patients · same site · same strength

$P(\text{yes} \mid q)$

Positive own effect

$W_{q,q} > 0$

Above controls

119 random + sham

$P(\text{yes} \mid q)$

Beats all 5 competitors

$W_{q,q} - W_{q,d} > 0$ for all $d \neq q$

All required; the own write must lead every competitor contrast.

Same Effusion question, different writes

Qwen2.5-VL-7B · NIH ChestX-ray14

Effusion direction → Effusion: $\Delta P(\text{yes}) = +0.190$

Nodule direction → Effusion: $\Delta P(\text{yes}) = +0.262$

random p95 = 0.137 · |sham| = 0.017

✓ Above controls

✗ Not owned: Nodule moves it more

$O_q = -0.071$